

Calculus

HW3

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Sec 1



Homework

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นายภิฑิตธิคุณ ทัดนพรมล

Find dy/dx by implicit differentiation.

$$y^2 = (x - y)(x^2 + y)$$

$$2y \cdot \frac{dy}{dx} = x^3 + xy - x^2y - y^2$$

$$2y \cdot \frac{dy}{dx} = 3x^2 + \left[x \frac{dy}{dx} + y \right] - \left[x^2 \frac{dy}{dx} + 2xy \right] - 2y \frac{dy}{dx}$$

$$= 3x^2 + x \frac{dy}{dx} + y - x^2 \frac{dy}{dx} - 2xy - 2y \frac{dy}{dx}$$

$$2y \frac{dy}{dx} - x \frac{dy}{dx} + x^2 \frac{dy}{dx} + 2y \frac{dy}{dx} = 3x^2 + y - 2xy$$

$$\frac{dy}{dx} [2y - x + x^2 + 2y] = 3x^2 + y - 2xy$$

$$\frac{dy}{dx} = \frac{3x^2 + y - 2xy}{x^2 - x + 4y}$$



$$y\sqrt{x} - x\sqrt{y} = 25$$

$$y(x)^{\frac{1}{2}} - x(y)^{\frac{1}{2}} = 25$$

$$\frac{y}{2\sqrt{x}} + \sqrt{x} \frac{dy}{dx} - \left(\frac{x}{2\sqrt{y}} \frac{dy}{dx} + \sqrt{y} \right) = 0$$

$$2\sqrt{xy} \left(\frac{2x \frac{dy}{dx} + y}{2\sqrt{x}} - \frac{2y + x \frac{dy}{dx}}{2\sqrt{y}} \right) = 2\sqrt{xy} \times 0$$

$$\sqrt{y} \left(2x \frac{dy}{dx} + y \right) - \sqrt{x} \left(2y + x \frac{dy}{dx} \right) = 0$$

$$2x\sqrt{y} \frac{dy}{dx} + y\sqrt{y} - 2\sqrt{x}y - x\sqrt{x} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} [2x\sqrt{y} - x\sqrt{x}] = 2\sqrt{x}y - y\sqrt{y}$$

$$\frac{dy}{dx} = \frac{2\sqrt{x}y - y\sqrt{y}}{2x\sqrt{y} - x\sqrt{x}}$$

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$$\cos(x + y) = x$$

$$-\sin(x+y) \cdot \left(1 + \frac{dy}{dx}\right) = 1$$

$$-\sin(x+y) - \sin(x+y) \frac{dy}{dx} = 1$$

$$-\sin(x+y) \frac{dy}{dx} = 1 + \sin(x+y)$$

$$\frac{dy}{dx} = \frac{1 + \sin(x+y)}{-\sin(x+y)}$$

$$\frac{dy}{dx} = - \frac{1 + \sin(x+y)}{\sin(x+y)}$$

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